

AI Impact on Jobs 2030 - Statistical Analysis

Notebook 2 : Statistical Analysis

Following the descriptive exploration of the AI Impact on Jobs 2030 dataset, this notebook delves deeper into **statistical analyses** to better understand patterns related to automation risk.

We focus on key job characteristics such as salary, experience, education, and AI exposure, and explore their relationships with automation probability. This includes examining distributions, testing statistical assumptions, and applying appropriate analyses to summarize and compare groups.

The main guiding question remains:

“Which types of jobs are most at risk of automation, and how do job characteristics are distributed across different levels of automation risk?”

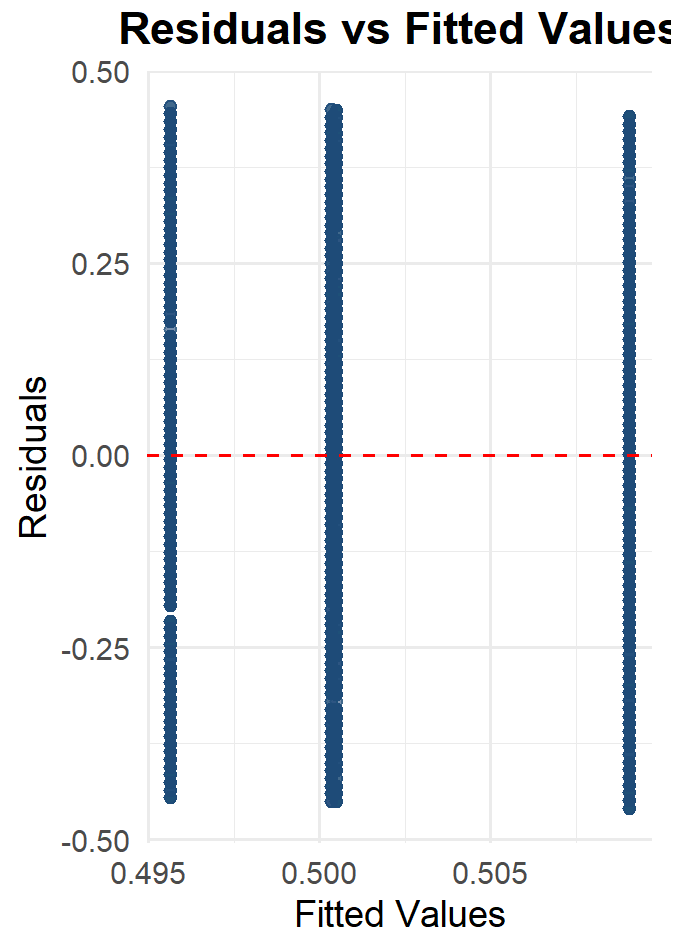
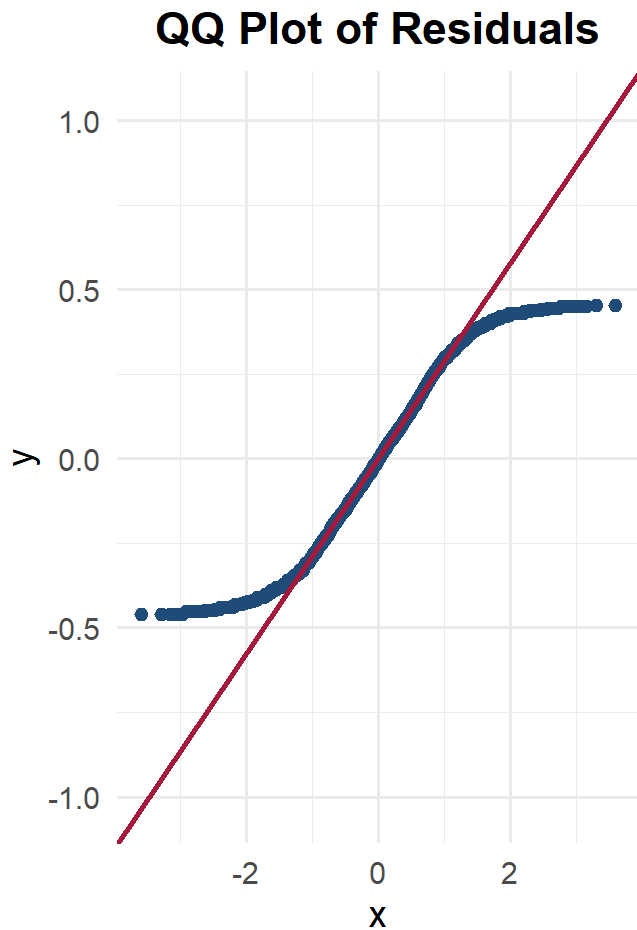
By rigorously examining these variables, we aim to provide clearer insights into the factors associated with automation vulnerability across occupations.

Author: J-F Jutras
Date: December 2025
Dataset: AI Impact on Jobs 2030 — Kaggle

Automation Probability Across Education Levels

ANOVA: Automation Probability by Education Level

term	Sum_Sq	Mean_Sq	F_value	P_value
Education_Level	0.072	0.024	0.389	0.761
Residuals	184.202	0.061	NA	NA



Normality and Homogeneity Tests for ANOVA

Test		Statistic	P_value
W	Shapiro-Wilk (Residuals)	0.967	0.000
...2	Levene (by Education Level)	1.595	0.189

A Kruskal-Wallis test was conducted in lieu of ANOVA due to a violation of the normality assumption in the residuals, whereas the assumption of homogeneity of variances was satisfied.

Kruskal-Wallis Test for Automation Probability by Education Level

	Statistic	df	p_value
Kruskal-Wallis chi-squared	1.2926	3	0.7309

Post-hoc Dunn Test (Bonferroni-adjusted) for Education Levels

Comparison	Z	P_unadj	P_adj	Significant
Bachelor's - High School	-0.7752	0.4382	1	No

Comparison	Z	P_unadj	P_adj	Significant
Bachelor's - Master's	0.3168	0.7514	1	No
High School - Master's	1.0860	0.2775	1	No
Bachelor's - PhD	-0.0054	0.9957	1	No
High School - PhD	0.7567	0.4492	1	No
Master's - PhD	-0.3169	0.7513	1	No

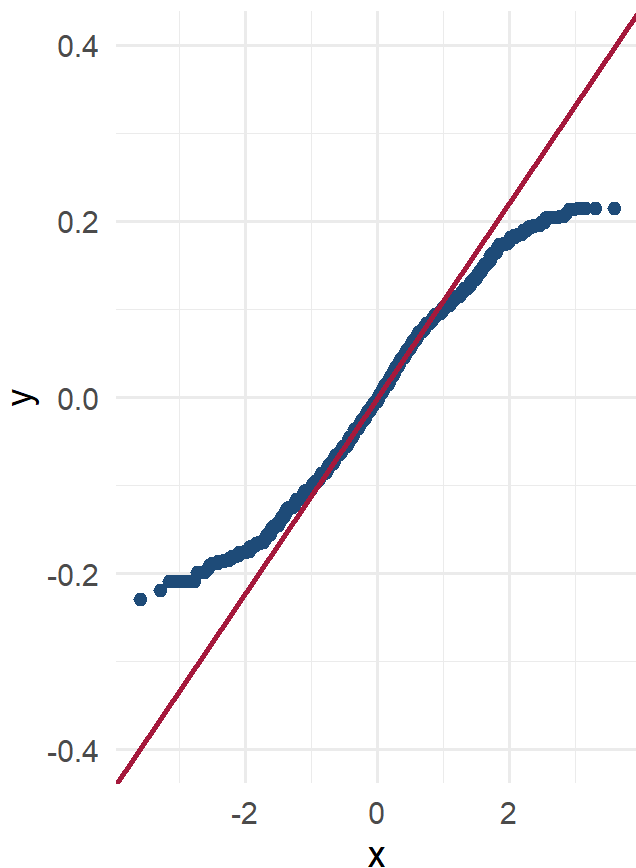
Both Kruskal-Wallis and Dunn tests confirm that automation probability does not significantly differ between education levels.

Automation Probability Across Job Title

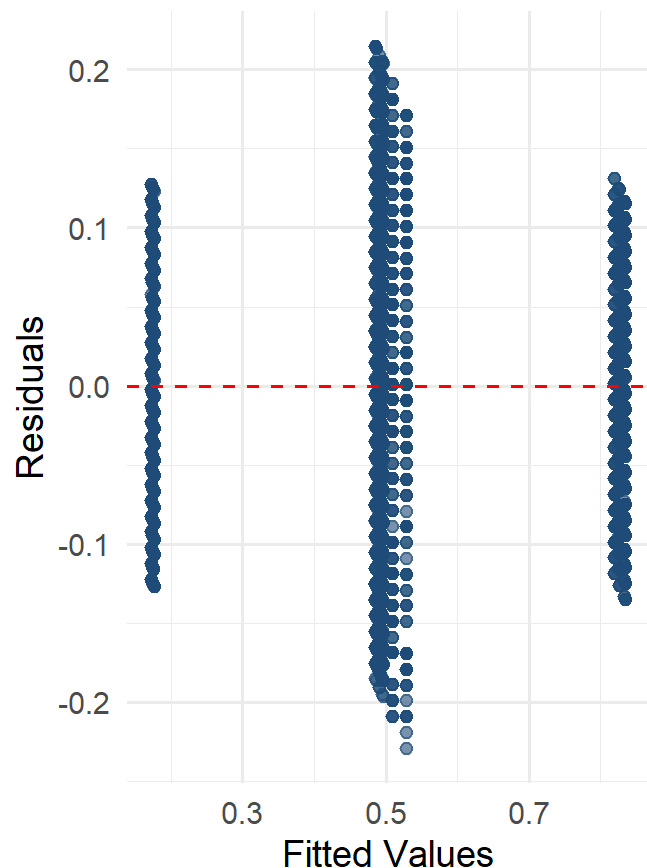
ANOVA: Automation Probability by Job Title

term	Sum_Sq	Mean_Sq	F_value	P_value
Job_Title	157.931	8.312	940.281	0
Residuals	26.343	0.009	NA	NA

QQ Plot of Residuals



Residuals vs Fitted Values



Normality and Homogeneity Tests for ANOVA

	Test	Statistic	P_value
W	Shapiro-Wilk (Residuals)	0.986	0
...2	Levene (by Job Title)	20.305	0

A one-way ANOVA revealed a highly significant effect of job title on automation probability but diagnostic tests indicated violations of the normality of residuals and homogeneity of variances. Consequently, a Welch ANOVA will be conducted to account for heteroscedasticity, and results will be complemented with appropriate post-hoc comparisons.

Welch ANOVA for Automation Probability by Job Title

	Statistic	df_num	df_den	p_value
F	1609.366	19	1088.927	0

Games–Howell Post-hoc Comparisons (Top 10)

.y.	group1	group2	estimate	conf.low	conf.high	p.adj	p.adj.signif
Automation_Probability_2030	AI Engineer	Chef	0.3235	0.2818	0.3652	0	****
Automation_Probability_2030	AI Engineer	Lawyer	0.3142	0.2724	0.3560	0	****
Automation_Probability_2030	AI Engineer	Marketing Manager	0.3226	0.2823	0.3628	0	****
Automation_Probability_2030	Chef	Construction Worker	0.3303	0.2894	0.3712	0	****
Automation_Probability_2030	Chef	Nurse	-0.3200	-0.3620	-0.2780	0	****
Automation_Probability_2030	Chef	Research Scientist	-0.3215	-0.3620	-0.2810	0	****
Automation_Probability_2030	Construction Worker	Mechanic	-0.3407	-0.3800	-0.3013	0	****
Automation_Probability_2030	Customer Support	Lawyer	-0.3388	-0.3801	-0.2976	0	****
Automation_Probability_2030	Doctor	Lawyer	0.3097	0.2679	0.3515	0	****
Automation_Probability_2030	Doctor	Mechanic	0.3086	0.2684	0.3487	0	****

A Welch ANOVA revealed a highly significant effect of job title on automation probability, $F(19,1088.93) = 1609.37$, $p < .001$, confirming substantial differences across occupations after accounting for heteroscedasticity.

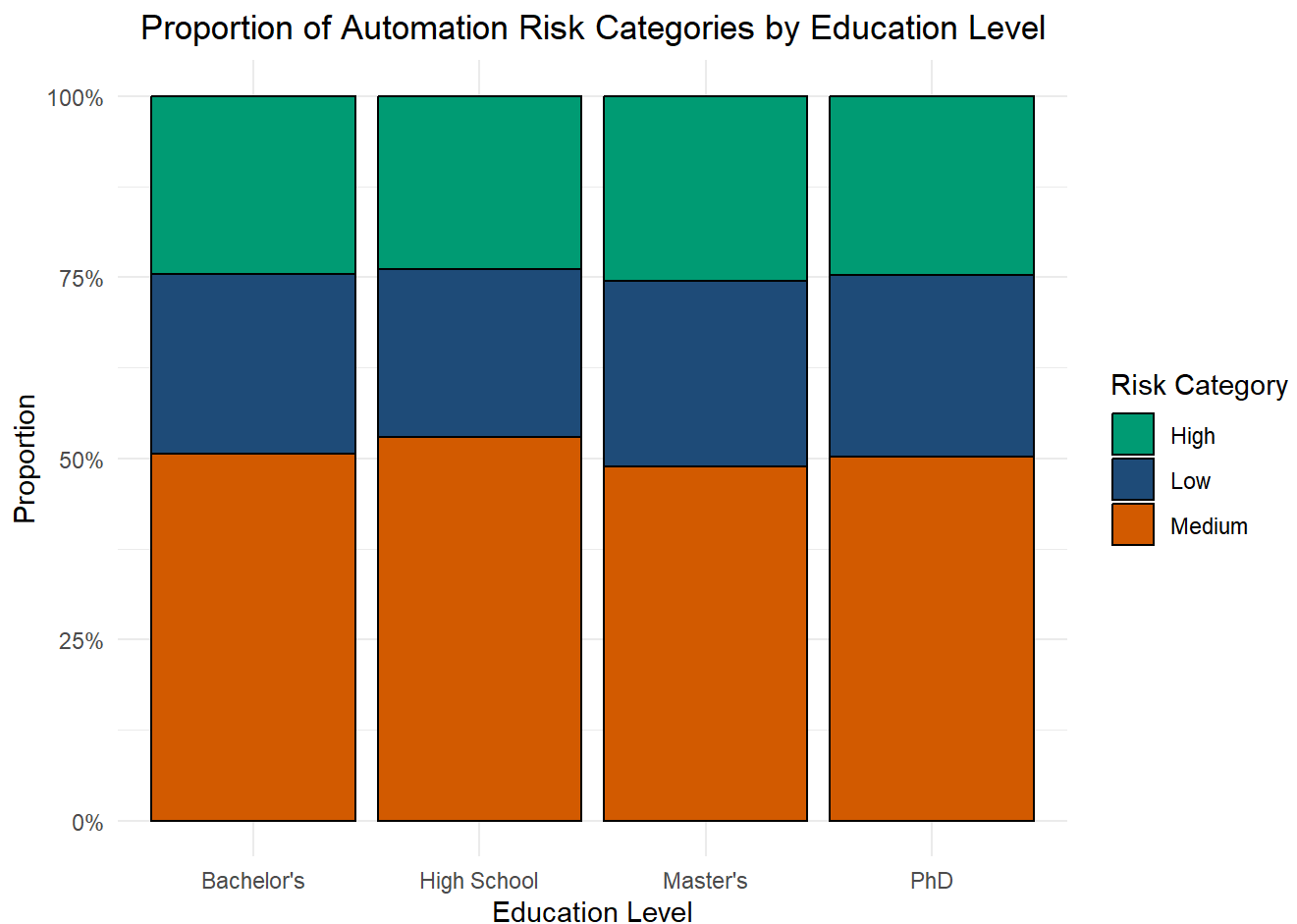
Games–Howell post-hoc comparisons indicated a clear occupational stratification of automation risk: highly skilled, knowledge-intensive professions exhibited similarly low automation probabilities, whereas routine, manual, and service-oriented occupations showed significantly higher levels of automation risk.

Together, these findings suggest that exposure to automation is primarily structured by broad occupational task characteristics rather than by individual job titles.

Risk Category vs. Education Level

Chi-Square Test for Education Level by Risk Category + Cramer's V

	Statistic	df	p_value	Cramers_V
X-squared	2.7192	6	0.8432	0.0213



The Kruskal-Wallis test showed no significant differences in automation probability across education levels. Consistently, the Chi-Square test on risk categories indicated no association with Education Level ($\chi^2 = 2.72$, $df = 6$, $p = 0.843$; Cramer's $V = 0.021$), suggesting that automation risk is largely independent of formal education.

Predictors of Automation Probability

Given the bounded nature of the dependent variable and the strong occupational clustering observed in preliminary analyses, a mixed-effects beta regression was estimated with job title specified as a random intercept. This specification allows individual-level predictors to be assessed net of occupational-level heterogeneity.

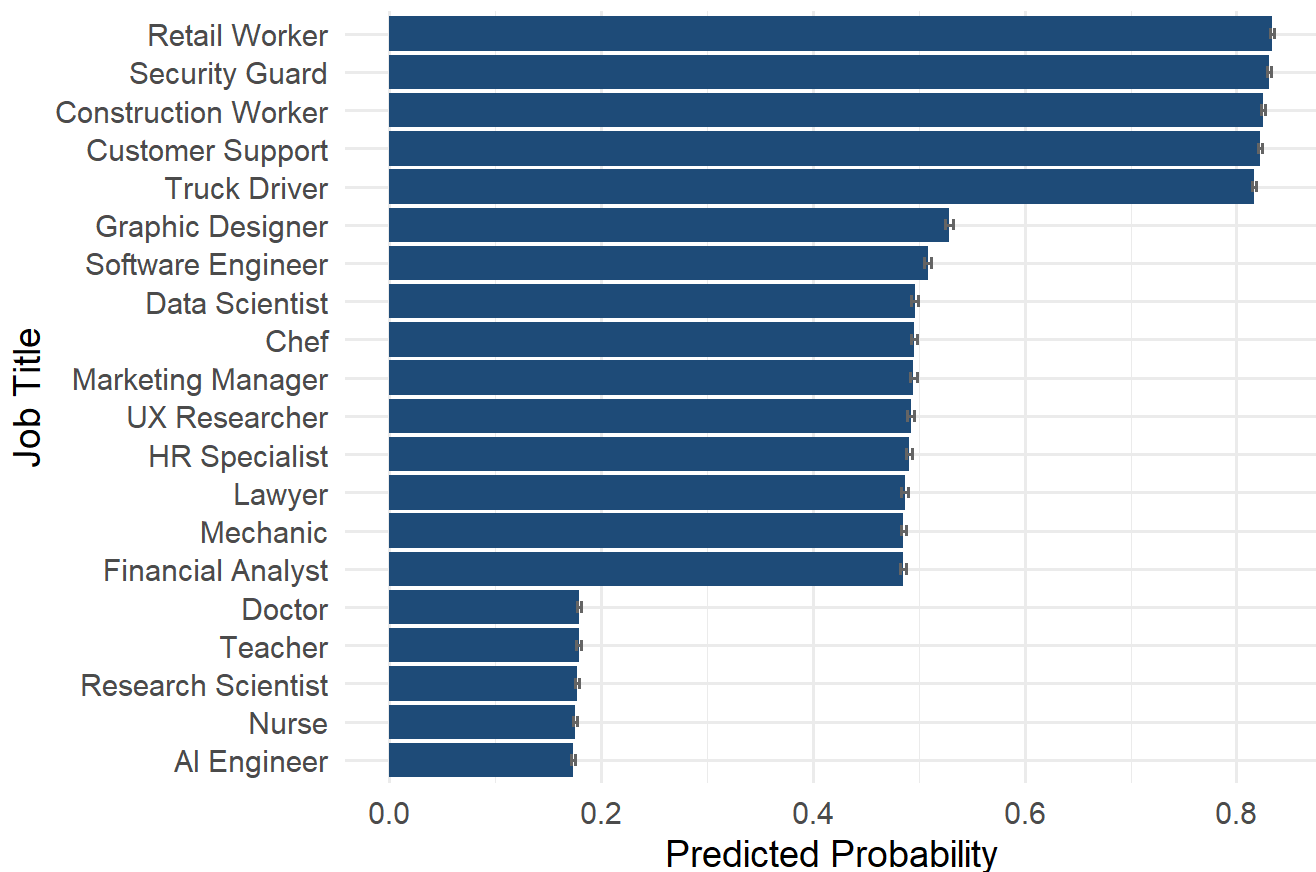
Mixed-Effects Beta Regression: Predictors of Automation Probability

	Estimate	Std_Error	z_value	p_value
(Intercept)	-0.0071	0.2489	-0.0285	0.9773
Average_Salary	0.0000	0.0000	0.3416	0.7327
Years_Experience	-0.0011	0.0010	-1.1707	0.2417
AI_Exposure_Index	-0.0150	0.0300	-0.5015	0.6160
Tech_Growth_Factor	0.0206	0.0296	0.6947	0.4872

A mixed-effects beta regression was estimated with job title specified as a random intercept to account for occupational-level heterogeneity. After controlling for this clustering, none of the individual-level predictors—average salary, years of experience, AI exposure, or technological growth—were statistically significant. Coefficient estimates were small and close to zero, indicating negligible marginal effects.

These findings suggest that variation in automation probability is largely driven by differences between occupations rather than by individual job characteristics within occupations.

Predicted Automation Probability by Job Title



This plot shows the predicted automation probability for each job title from the mixed-effects Beta regression model, with bars representing the mean predicted probability and error bars indicating variability across individuals within each job. Jobs are ordered from lowest to highest risk, highlighting which occupations are most and least exposed to automation.

Key Takeaways: Statistical Analysis

- **Education does not explain automation risk:** Non-parametric tests (Kruskal-Wallis, Dunn) and Chi-Square analyses show no significant differences in Automation Probability or Risk Categories by Education Level. The interaction between AI Exposure and Education is also non-significant, reinforcing the minimal role of formal education.
- **Job title drives differences in automation risk:** Welch ANOVA and Games-Howell post-hoc tests reveal substantial variation in Automation Probability across occupations. Highly skilled, knowledge-intensive jobs show low automation risk, whereas routine/manual/service-oriented jobs show high risk.
- **Continuous predictors show negligible effects:** Mixed-effects Beta regression with job title as a random intercept indicates that Average Salary, Years of Experience, AI Exposure, and Tech Growth have small, non-significant effects on automation probability, highlighting that individual-level predictors explain little beyond occupational clustering.
- **Automation risk is structured by occupation, not education:** The combination of statistical tests confirms that differences in automation exposure are primarily driven by job-level characteristics rather than by individual attributes such as education or experience.